Progressive Microcytic Anemia

CASE 02 | FICTIONAL PERSON | NO REAL PATIENT DATA

Synthetic model-evaluation document. Not for clinical use. Do not use these values to make decisions about a real person.

Clinical note

A 58-year-old man reports progressive fatigue and reduced exercise tolerance over five months. He denies visible blood in stool or urine, recent infection, fever, or unintentional weight loss. No prior hematologic diagnosis is documented.

Examination note: he is hemodynamically stable. Mild conjunctival pallor is noted. No jaundice, lymphadenopathy, or palpable abdominal mass is recorded.

Relevant history	No chronic kidney disease, inflammatory bowel disease, bariatric surgery, or known hemoglobinopathy.
Medication	No anticoagulant use. No iron supplement use before testing.
Diet and blood loss	Diet is mixed. No blood donation in the last year. One fecal immunochemical test is negative.
Prior CBC	Hemoglobin was 12.6 g/dL twelve months earlier; MCV was 84 fL.

All names, dates, histories, values, intervals, and notes in this case are fabricated.

Complete blood count and chemistry

The current CBC shows anemia with small, hypochromic red cells. The platelet count is mildly above the local interval.

Test	Result	Unit	Printed reference interval	Report flag
Hemoglobin	9.4	g/dL	13.5 - 17.5	L
Hematocrit	30.1	%	41 - 53	L
RBC count	4.27	10^12/L	4.50 - 5.90	L
MCV	70.5	fL	80 - 100	L
MCH	22.0	pg	27 - 33	L
RDW	18.2	%	11.5 - 14.5	H
Platelets	442	10^9/L	150 - 400	H
Creatinine	0.91	mg/dL	0.70 - 1.30	

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Iron studies and inflammatory markers

Iron studies were collected with the CBC. There is no clinical or laboratory evidence of an acute inflammatory illness in this sample.

Test	Result	Unit	Printed reference interval	Report flag
Ferritin	6	ng/mL	30 - 400	L
Serum iron	20	mcg/dL	65 - 175	L
TIBC	442	mcg/dL	250 - 450	
Transferrin saturation	5	%	20 - 50	L
CRP	1.2	mg/L	< 5.0	
Reticulocyte count	0.8	%	0.5 - 2.5	
Blood smear	Microcytosis and hypochromia			

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